

# MENGM0056 - Product and Production Systems

## Scenario 3: FMCG - Bottled Beverage (500 ml)

Hand-out for Group Coursework (2025/26)

**UUID seed:** 977795e2-63d3-482e-bc32-72954b136416    **Checksum:** 6292bbe21cfd

### **Purpose**

This scenario considers a high-throughput beverage line with volatile demand and despatch congestion. Your task is to propose operational policies that stabilise service level and improve utilisation while controlling changeover losses and inventory.

### **Narrative**

A 500 ml carbonated soft drink is produced in PET bottles. The line comprises blow-moulding, filling, labelling, case-packing and palletising, with despatch to outbound trucks via limited loading bays. Demand varies with weather and promotions. CIP and changeovers consume valuable capacity. Capital spend is constrained; improvements should focus on scheduling, policies, and parameter changes.

### **Entities and flow (fixed structure)**

Preforms → Blow-mould → Fill → Cap → Label → Case-pack → Palletise → Despatch.

## Baseline parameters (seeded)

### Global

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|                         |                                  |
|-------------------------|----------------------------------|
| Shifts per day          | 2                                |
| Shift length            | 7.5 h                            |
| Base daily demand       | 2251 cases/day (12 bottles/case) |
| Daily demand CV         | 0.325                            |
| Number of SKUs          | 5                                |
| On-time despatch target | 95%                              |

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### Line capacities and availability

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| Resource     | Count | Nominal rate | Availability |
|--------------|-------|--------------|--------------|
| Blow-moulder | 1     | 24235 bph    | 0.888        |
| Filler       | 1     | 20099 bph    | 0.832        |
| Labeller     | 1     | 23188 bph    | 0.89         |
| Case-packer  | 1     | 1996 cph     | 0.893        |
| Palletiser   | 1     | 1928 cph     | 0.916        |

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### Changeovers and CIP

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|                                      |           |
|--------------------------------------|-----------|
| CIP duration (flavour)               | 49 min    |
| Additional flavour change operations | 28 min    |
| Label-only change duration           | 9 min     |
| Minimum batch size                   | 488 cases |

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## Despatch and yard logistics

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|                          |              |
|--------------------------|--------------|
| Loading bays             | 3            |
| Despatch window          | 7:00 – 18:00 |
| Mean truck inter-arrival | 45 min       |
| Truck service time       | 51 min       |
| Cases per pallet         | 108          |
| Pallets per truck        | 27           |

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## Reliability (downtime parameters)

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| Resource    | MTBF (min) | MTTR (min) |
|-------------|------------|------------|
| BlowMoulder | 419.3      | 22.4       |
| Filler      | 393.4      | 27.9       |
| Labeller    | 879.7      | 18.7       |
| Packer      | 926.8      | 23.7       |
| Palletiser  | 836.1      | 17.7       |

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## Costs

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|                             |                     |
|-----------------------------|---------------------|
| Holding cost                | £1.42 /pallet/day   |
| Changeover cost (all-in)    | £254.77 /event      |
| Lateness penalty            | £274.45 /late truck |
| Scrap cost (changeover/CIP) | £1.8 /case          |

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## Required KPIs

- Line utilisation by unit (blow-moulder, filler, labeller, packer, palletiser).
- Changeover time and product loss per week; percentage of capacity lost to changeovers/CIP.
- Order lead time distribution and on-time despatch rate (service level).
- Loading-bay utilisation and maximum truck queue length; truck lateness count.
- Finished-goods days-of-cover and average pallets in buffer.

## Techniques to apply

- **Modelling & KPIs:** capacity model, bottleneck identification, changeover loss accounting.
- **Mathematical programming:** shift patterns, SKU sequencing and batch sizing subject to CIP and bay constraints.
- **Uncertainty modelling:** daily demand and truck arrivals; downtime distributions.
- **Simulation:** discrete-event model of the line and despatch yard; evaluate congestion and schedules.
- **Metaheuristic optimisation:** lot-sizing and sequence optimisation with changeover penalties and service-level targets.

## Improvement levers (examples)

- SKU sequencing to group labels and reduce full CIP events; threshold policies for label-only changes.
- Time-of-day despatch smoothing: reserve windows for large orders; dynamic bay assignment.
- Buffer targets before palletiser and before despatch to prevent starvation/blocking.
- Preventive maintenance windows aligned with expected demand troughs.

## Deliverables

1. A report (max 20 sides of A4 including figures and references; appendices unmarked but admissible as evidence).
2. The report should contain a production and despatch plan for one representative week, showing SKU sequence, batch sizes, and expected service level.
3. Model files (e.g., simulation, optimisation) as appendices/evidence.

## Assessment emphasis

Clarity and correctness of the capacity and KPI model; appropriate choice and justification of techniques; quality of experimental design; robustness to demand variability; and persuasiveness of recommendations under operational constraints.

## Data ethics and reproducibility

Report your UUID seed and any random seeds used within tools. Provide enough detail for independent regeneration of your parameter tables.